Delegating a contract to a planner in series with an existing controller

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1 Setting

Ports and behaviours. The ports are $\Pi = U \cup W \cup E \cup Y$: U is the planner's input (fence, home, goal), W the wire (a mission: at most N items, each an item kind and a waypoint, stored as a fixed-length vector with a length field), $E = E_s \cup E_d$ the environment (static part E_s: start position, parameters; per-tick part E_d: measured dt, estimates), Y the controller's per-tick targets. A behaviour assigns a value to every port at each tick $t \in \{0, \dots, H\}$; ports in U, W, E_s keep one value over the horizon, which removes the mismatch between a finite list and per-tick signals. Ω is the set of behaviours and $\neg S := \Omega \setminus S$.

Projections. For $X \subseteq \Pi$ and $S \subseteq \Omega$, $\exists_X S$ is the set of behaviours that agree outside X with some member of S, and $\forall_X S := \neg\exists_X \neg S$. S is *over* $Z \subseteq \Pi$ if $\exists_{\Pi \setminus Z} S = S$. We use: (F0) sets over Z are closed under $\neg, \cap, \cup$ and are over every $Z' \supseteq Z$; (F1) $\forall_X S \subseteq S \subseteq \exists_X S$, and both operators are monotone; (F2) if S is over Z then $\exists_X S$ and $\forall_X S$ are over $Z \setminus X$; (F3) if C is over $\Pi \setminus X$ then $C \subseteq S$ iff $C \subseteq \forall_X S$, and $S \subseteq C$ iff $\exists_X S \subseteq C$; (F4) if C is over $\Pi \setminus X$ then $\exists_X (C \cap S) = C \cap \exists_X S$.

Contracts. A contract is a pair (A, G) of subsets of Ω, and $M \models (A, G)$ iff $M \cap A \subseteq G$. Given: A over $U \cup E$ and G over $U \cup E \cup Y$, both PolySTL formulas (atoms $p \geq 0$ or $p > 0$, p a rational polynomial in the values at the current tick; $\neg, \wedge, \vee$; $\Box_I, \Diamond_I, \mathcal{U}_I$ with integer windows $I \subseteq [0, H]$); the controller M_2 over $W \cup E \cup Y$, given as deterministic code with bounded loops. The planner M_1 over $U \cup W$ does not exist yet. Wanted: A_1 over U, G_1 over $U \cup W$, A_2 over $W \cup E$, G_2 over $W \cup E \cup Y$ with (D1) $A \subseteq A_1$, (D2) $G_1 \subseteq A_2$ and (D3$^+$) $A \cap G_1 \cap A_2 \cap G_2 \subseteq G$.

Why D3$^+$ and not $A \cap A_2 \cap G_2 \subseteq G$. $A_2 \cap G_2$ is over $\Pi \setminus U$, so by F3 the shorter form is equivalent to $A_2 \cap G_2 \subseteq \forall_U (\neg A \cup G)$: the controller alone would have to keep G for every planner input at once (for square fences of every half-side $s > 0$ this pins the target to the origin). Only G_1 can relate u to the mission, so the third condition must contain it; D3$^+$ is implied by the shorter form and by $G_2 \cap A_2 \subseteq G$, so it loses nothing. Likewise, if A_2 kept the environment conditions P of the code, D2 would force $G_1 \subseteq \forall_E A_2$, which is empty once P bounds dt; step 2 moves P into G_2, which leaves the controller's obligation unchanged (Theorem, d).

Pointwise class. Φ consists of formulas $\beta \wedge \Box_{[0,H]} \alpha \wedge \bigwedge_{j=1}^{m} \Diamond_{[0,H]} \gamma_j$ with β quantifier-free over the static ports $U \cup W \cup E_s$ and α, γ_j quantifier-free over static ports and the values of one tick; we set $\gamma_0 := \top$.

2 Algorithm

Input: A, G, H, the code of M_2. *Output:* (A_1, G_1), (A_2, G_2) and a set *Bad* over U, or the report "no controller guarantee".

1. Controller vocabulary (reads M_2 only). From the guards and clamps of M_2 read three PolySTL formulas: V over W (accepted item kinds, sentinel values excluded, length at most N, leg lengths in $[\ell_{\min}, \ell_{\max}]$), P over $W \cup E$ (ranges of dt and parameters, leg conditions that involve the start position; P may also take over the conjuncts of A that mention only E, since A lies inside them), and a chain $T_1 \subseteq \cdots \subseteq T_k$ over $W \cup E \cup Y$, where T_j says that at every tick the target lies within ρ_j of the polyline from the start through the waypoints and that some tick brings it within η of the last waypoint. ρ_1 is the largest distance to the polyline observed when M_2 is run on a fixed finite list of legs with nominal inputs (no valid tube can be thinner), and $\rho_j := \rho_1 + (j-1)\kappa$ for a fixed step κ and length k.

2. Certify (reads M_2 only). Decide $M_2 \cap V \cap P \subseteq T_j$ by bounded model checking of the code unrolled over the horizon $[0, H]$ of A and G (a guarantee certified on a shorter horizon says nothing about the remaining ticks and would empty G_1). The verified indices form an up-set of the chain, so binary search finds the least verified j^* in $\lceil \log_2 k \rceil + 1$ checks. If there is none, stop with "no controller guarantee". Set $A_2 := V$ and $G_2 := \neg P \cup T_{j^*}$.

3. Planner assumption. $A_1 := \exists_{E \cup Y} A$.

4. Planner guarantee. $R := \forall_{E \cup Y} (\neg A \cup \neg A_2 \cup \neg G_2 \cup G)$ and $G_1 := A_2 \cap R$, computed as in Section 5.

5. Realizability. $Bad := A_1 \cap \neg\exists_W G_1$. Return both contracts and *Bad*, and report a vacuous query if $A_1 = \emptyset$. If $Bad = \emptyset$, every planner input allowed by A has a mission meeting G_1; otherwise *Bad* is exactly the set of inputs that no mission serves under this controller guarantee, and no other certified member of the chain gives a smaller *Bad* (Theorem, e). A further repair needs a thinner tube than any certified one; the strongest guarantee M_2 admits under V is $M_2 \cap V$ itself, i.e. the code inlined, which the method excludes.

Every step is deterministic: the chain, the test legs, the variable orders and the search orders are fixed, and the decision procedures below are deterministic. A guarantee for M_2 that mentioned U would gain nothing: $M_2 \cap A_2$ is over $\Pi \setminus U$, so by F3 it lies in G_2 iff it lies in $\forall_U G_2$.

3 Theorem

(a) A_1 is over U, G_1 over $U \cup W$, A_2 over W, G_2 over $W \cup E \cup Y$. (b) D1, D2 and D3$^+$ hold. (c) If $M_1 \models (A_1, G_1)$ and $M_2 \models (A_2, G_2)$ then $M_1 \cap M_2 \models (A, G)$ and $\exists_W (M_1 \cap M_2) \models (A, G)$. (d) $M_2 \models (A_2, G_2)$ iff $M_2 \cap V \cap P \subseteq T_{j^*}$, which step 2 established. (e) Every G_1' over $U \cup W$ with $G_1' \subseteq A_2$ and $A \cap G_1' \cap A_2 \cap G_2 \subseteq G$ satisfies $G_1' \subseteq G_1$, and among the certified members of the chain T_{j^*} yields the largest G_1. (f) A planner M_1 that answers every input ($\exists_W M_1 = \Omega$) with $M_1 \models (A_1, G_1)$ exists iff $Bad = \emptyset$.

Hence $G_1 \neq \emptyset$ iff some G_1' meeting D2 and D3$^+$ for the chosen (A_2, G_2) is nonempty, iff $V \cap \forall_{E \cup Y} (\neg A \cup \neg G_2 \cup G) \neq \emptyset$; when every (u, e) admits a target outside G, this reads: some mission w in V and input u have $A \subseteq P$ and $A \cap T_{j^*} \subseteq G$ at (u, w). G_1 is built from A, G, V, P, T_{j^*} only, so its size does not depend on the code of M_2.

4 Proof

(a) A is over $U \cup E$, so A_1 is over U by F2. The set inside R is over Π (F0), so R is over $U \cup W$ by F2; V is over W, so G_1 is over $U \cup W$ by F0; G_2 is over $W \cup E \cup Y$ by F0.

(b) D1 is F1. D2: $G_1 = A_2 \cap R \subseteq A_2$. D3$^+$: by F1, $A \cap G_1 \cap A_2 \cap G_2 \subseteq (A \cap A_2 \cap G_2) \cap (\neg(A \cap A_2 \cap G_2) \cup G) \subseteq G$.

(c) Let $M := M_1 \cap M_2$. By D1, $M \cap A \subseteq M_1 \cap A_1 \subseteq G_1$; by D2, $M \cap A \subseteq M_2 \cap G_1 \subseteq M_2 \cap A_2 \subseteq G_2$; so $M \cap A \subseteq A \cap G_1 \cap A_2 \cap G_2 \subseteq G$

by D3$^+$. A and G are over $\Pi \setminus W$, so F4 gives $\exists_W M \cap A = \exists_W (M \cap A)$ and F3 gives $\exists_W (M \cap A) \subseteq G$.

(d) $M_2 \cap V \subseteq \neg P \cup T_{j^*}$ iff $M_2 \cap V \cap P \subseteq T_{j^*}$.

(e) $A \cap G_1' \cap A_2 \cap G_2 \subseteq G$ iff $G_1' \subseteq \neg(A \cap A_2 \cap G_2) \cup G$, iff $G_1' \subseteq R$ by F3, as G_1' is over $\Pi \setminus (E \cup Y)$; with $G_1' \subseteq A_2$ this gives $G_1' \subseteq G_1$. If $T \subseteq T'$ then $\neg P \cup T \subseteq \neg P \cup T'$, so the R built from T' is contained in the R built from T (F1).

(f) If $Bad = \emptyset$, take $M_1 := G_1 \cup \neg A_1$: $\exists_W M_1 \supseteq \exists_W G_1 \cup \neg A_1 \supseteq A_1 \cup \neg A_1 = \Omega$ (F1 and $Bad = \emptyset$) and $M_1 \cap A_1 \subseteq G_1$. Conversely, by F4 and F1, $A_1 = A_1 \cap \exists_W M_1 = \exists_W (A_1 \cap M_1) \subseteq \exists_W G_1$. □

5 Decidability and cost

Termination. Five steps, a finite chain, and each call below terminates.

Step 2 is the only step that reads code, and it reads M_2 alone. With bounded loops and IEEE arithmetic the code unrolled over H ticks is a finite circuit on finitely many bit-vector inputs; V, P, T_j unroll to finitely many polynomial atoms on those inputs, so the inclusion is a finite satisfiability question (bit-precise SMT; real quantifier elimination if the code is read over the reals). Running the code (step 1) only bounds ρ from below; with measured dt and estimates as free inputs a finite set of runs certifies nothing, so runs never replace step 2. Symmetry of the code under rigid motions is not exact under rounding and is not relied on. A deadline of D seconds becomes the tick window $[0, \lfloor D/\overline{dt} \rfloor]$, with $\overline{dt}$ the largest tick length A admits; this is sound because no tick is longer, and its loss is the ratio of the tick-length bounds, which is why P should take over the range of dt from A.

Steps 3–5 never read code. Unrolling over $[0, H]$ turns A, G, V, P, T_{j^*} into quantifier-free formulas of real arithmetic over the samples; $\exists_X$ and $\forall_X$ become blocks of real quantifiers over the samples of X, and real quantifier elimination (cylindrical algebraic decomposition; Tarski–Seidenberg) returns an equivalent quantifier-free polynomial formula over the static samples of $U \cup W$, that is a PolySTL formula without temporal operators read at tick 0; $Bad = \emptyset$ is a closed sentence decided the same way. This is complete but doubly exponential in $(H+1)(|E_d|+|Y|)$. (If U were per-tick, an output atom could couple two ticks, which is outside PolySTL; the output then stays in quantifier-free real arithmetic over samples.)

Pointwise lemma. Let $\varphi = \beta \wedge \Box\alpha \wedge \bigwedge_{j \le m} \Diamond\gamma_j \in \Phi$ with $H \ge m$. Over all signals, $\varphi \to \Box c$ holds iff $\neg\beta$, or $\alpha \cap \gamma_j = \emptyset$ for some $j \le m$, or $\alpha \subseteq c$; and $\varphi \to \Diamond d$ holds iff $\neg\beta$ or $\alpha \cap \gamma_j \subseteq d$ for some $j \le m$ (sets of one tick's values). Reason: witnesses of the $\Diamond\gamma_j$ go on distinct ticks and every other tick repeats one of them or carries the value that breaks c, so a violating signal exists iff violating tick values do; $H \ge m$ is needed (it fails for smaller H). When A, P, T_{j^*}, G are in Φ, step 4 splits $\neg P$ and the conjuncts of G and calls quantifier elimination over one tick's $E_d \cup Y$ values and E_s, independently of H. For convex polygonal fences and tubes no solver is needed: the largest value of $n \cdot z$ over the ρ-neighbourhood of a segment $[a, b]$ is $\max(n \cdot a, n \cdot b) + \rho\|n\|$.

Cost. A monolithic check decides, for one planner, one query on the unrolled code of M_1 and M_2 together, and is repeated for every planner revision. Here every decision procedure sees one component's artefacts: step 2 sees the unrolled M_2 and the formulas V, P, T_j, of size independent of G; steps 3–5 see only formulas; in Φ each call eliminates $|E_s| + |E_d| + |Y|$ variables, independent of H and of the code. Each planner is then checked against G_1 alone, a quantifier-free formula over static $U \cup W$: no unrolling of M_2 and no per-tick variable. Since the unrolled M_2 (about $1.2 \cdot 10^4$ ticks for a 30 s mission at the rate A admits in Section 6) dominates both encodings and is shared, the delegation costs about one controller check plus $O(\log k)$ repeats of it, done once, and every planner check is far smaller than a monolithic one.

6 Worked example

Data. $u = (s, h, g)$: square fence $F_s = [-s, s]^2$, home $h \in \mathbb{R}^2$, goal $g \in \mathbb{R}^2$. $E_s = \{p_0\}$ (start position), $E_d = \{dt\}$; $Y = \{y\}$, $y_t \in \mathbb{R}^2$ the horizontal target; $W = (w_1, w_2)$, two straight legs $p_0 \to w_1 \to w_2$. Write $\|z\|_\infty := \max(|z_x|, |z_y|)$ (so $\|z\|_\infty \le c$ and $z \in F_s$ are four linear atoms each), $q \times r := q_x r_y - q_y r_x$, and $S_\rho(a, b)$ for the points within ρ of $[a, b]$, the polynomial disjunction $\|z - a\|^2 \le \rho^2 \vee \|z - b\|^2 \le \rho^2 \vee \big(0 \le (z - a) \cdot (b - a) \le \|b - a\|^2 \wedge ((z - a) \times (b - a))^2 \le \rho^2 \|b - a\|^2\big)$. $A := s > 0 \wedge \|p_0 - h\|^2 \le \delta^2 \wedge \Box(0.0024 \le dt \le 0.0026)$ and $G := \Box(y \in F_s) \wedge \Diamond(\|y - g\|^2 \le \varepsilon^2)$ with $\delta = 1$, $\varepsilon = 3$. Suppose steps 1–2 give $V := \ell_{\min}^2 \le \|w_2 - w_1\|^2 \le \ell_{\max}^2$, $P := \ell_{\min}^2 \le \|w_1 - p_0\|^2 \le \ell_{\max}^2 \wedge \Box(\frac{1}{2000} \le dt \le \frac{1}{50})$ and $T := \Box\big(y \in S_\rho(p_0, w_1) \cup S_\rho(w_1, w_2)\big) \wedge \Diamond(\|y - w_2\|^2 \le \eta^2)$ with $\ell_{\min} = 5$, $\ell_{\max} = 100$, $\rho = 2$, $\eta = 0.5$ (illustrative values; the tube width covers the corner blend and half a tick of travel).

Step 3. $A_1 = (s > 0)$. **Step 4 by hand**, for $s > 0$, in four parts. (i) $\neg P$ through dt: $\Box(dt \in [0.0024, 0.0026]) \wedge \Diamond(dt \notin [\frac{1}{2000}, \frac{1}{50}])$ has $\alpha \cap \gamma_1 = \emptyset$, so it is vacuous. (ii) $\neg P$ through the first leg: some target breaks G, so every start allowed by A must meet the leg condition: $(\ell_{\min} + \delta)^2 \le \|w_1 - h\|^2 \le (\ell_{\max} - \delta)^2$. (iii) T and the fence, by the lemma ($c = F_s$) and support functions over all p_0 in the disc: $\|h\|_\infty \le s - \rho - \delta$, $\|w_1\|_\infty \le s - \rho$, $\|w_2\|_\infty \le s - \rho$. (iv) T and the goal, by the lemma (d the ε-disc at g; $\alpha \cap \gamma_1$ is the η-disc at w_2, which lies in the tube as $\eta \le \rho$): $\|w_2 - g\|^2 \le (\varepsilon - \eta)^2$. So $G_1 = V \wedge \big(s \le 0 \vee [(\text{ii}) \wedge (\text{iii}) \wedge (\text{iv})]\big)$, polynomial atoms over $U \cup W$ only, found without a solver.

Instance $s = 50$, $h = (-40, -40)$, $g = (40, 40)$. The mission $((-30, 30), (40, 40))$ meets G_1 (legs 70.7 and 70.7, coordinates within 48, $w_2 = g$). $((0, -49), (40, 40))$ breaks $\|w_1\|_\infty \le 48$: the tube contains $(-1.72, -50.02)$, outside F_{50}; the real controller cuts this corner inward, which G_1 cannot see without inlining M_2. $((-30, 30), (44, 40))$ breaks the goal part. Any w_1 within 6 m of h is rejected: some start allowed by A then makes the first leg shorter than $\ell_{\min}$, which P excludes. **Step 5** at $s = 50$: $Bad = \{\|h\|_\infty > 47\} \cup \{\text{dist}(g, [-48, 48]^2) > 2.5\}$; otherwise $w_2 :=$ the point of $[-48, 48]^2$ nearest g and $w_1 :=$ the midpoint of h, w_2 (or, when $\|h - w_2\| < 12$, any point of $[-48, 48]^2$ at distance 20 from h) meet G_1. An exhaustive search over extreme starts and targets on $5 \cdot 10^4$ random missions agreed with G_1 on every one, and for 25 fixed missions meeting G_1 an SMT solver found no start, tick lengths or targets violating G (unrolled over two ticks), while the symbolic query in all of u, w did not finish in 10 minutes.